

ABSTRACT

The project “**Real-Time Oscilloscope and Its Application Using Arduino**” focuses on developing a low-cost, portable oscilloscope that can display real-time electrical signals using an Arduino microcontroller and a TFT LCD screen. Traditional oscilloscopes, though accurate and feature-rich, are expensive and not easily accessible for students or small laboratories. This project presents an affordable alternative by utilizing the analog-to-digital conversion (ADC) capability of the Arduino to sample analog signals and display them graphically in real time. The input analog signal, such as a varying voltage from a function generator, is fed into the Arduino’s analog input pin (A15). The Arduino continuously reads this signal, processes the data, and transmits it to the TFT display. The display then plots the waveform corresponding to the voltage variation over time, creating a visual representation similar to a commercial oscilloscope. The system refreshes rapidly, allowing users to observe changing signal patterns almost instantaneously. The project finds applications in educational labs, circuit testing, sensor monitoring, and DIY electronics experiments, providing a hands-on understanding of signal behaviour and microcontroller interfacing.

CHAPTER 1

PREAMBLE

1.1 GENERAL INTRODUCTION

The oscilloscope is one of the most powerful electronic instruments that is available to electronics hobbyist, experimenters, and engineers. Oscilloscopes are used to observe the change of an electrical signal overtime, such that voltage and time describes a shape which is continuously graphed against a calibrated scale. It is mainly used to measure time-varying signals. Due to the size, the cost and complex structure of oscilloscope there is need to alternative solution.

In order to solve this problem the creation of a cheaper oscilloscope will be vital. This project aims to construct a low cost oscilloscope using readily available components like an Arduino microcontroller and a TFT display. By leveraging the versatility of Arduino and the visual capabilities of the TFT screen, the project strives to create a real-time visualization tool for various electronic signals.

Arduino is programmed and interfaced with TFT Module to plot the graph for outputs from the serial monitor of Arduino or TFT screen. Here Arduino is programmed such that it displays the output waveforms as the oscilloscope does. Arduino gives values only, so it is interfaced with TFT Module using graphics library are plotted for the outputs from Arduino plotter on TFT screen. This oscilloscope can be used to measure the frequency up to 100 KHz. The ARDUINO Board, the heart of the oscilloscope read value from its inbuilt analog to digital converter (ADC) and pushes this to TFT module by USB port.

Furthermore, the project incorporates a frequency generator as a controlled signal source. This enables the testing and calibration of the built in oscilloscope functionality, ensuring accurate signal representation and providing a reliable reference point for further measuring the experimenting with different signals frequencies generated by the frequency generator that aims to showcase the versatility and adaptability of the developed oscilloscope.

1.2 OBJECTIVES

The main objective of this project is to design and develop a low-cost real-time oscilloscope using an Arduino microcontroller and a TFT LCD display to visualize analog electrical signals. The project aims to create an affordable and portable device that can display the waveform of voltage signals in real time, similar to a conventional oscilloscope. It seeks to utilize the Arduino's Analog-to-Digital Converter (ADC) to convert analog signals into digital data, which can then be processed and graphically represented on the screen. Another important goal is to provide a user-friendly interface for easy signal observation and analysis, especially for students and electronics enthusiasts. Ultimately, the project aims to demonstrate that a simple and cost-effective Arduino-based setup can serve as a practical alternative to expensive commercial oscilloscopes for basic circuit testing and educational applications.

1.3 EXPECTED OUTCOMES:

1.3.1 SUCCESSFUL REAL-TIME SIGNAL DISPLAY:

The oscilloscope is expected to accurately capture and display analog voltage signals in real time on the 2.4-inch TFT LCD screen, representing waveform variations clearly over a time axis.

1.3.2 LOW-COST AND PORTABLE OSCILLOSCOPE PROTOTYPE:

The project aims to deliver an affordable and compact alternative to traditional laboratory oscilloscopes, making it suitable for students, hobbyists, and small-scale electronics testing.

1.3.3 FUNCTIONAL GRAPHICAL INTERFACE:

A well-structured graphical interface with grid lines, voltage labels, and time scaling will enhance user readability and allow easier interpretation of signal behaviour.

1.3.4 ADJUSTABLE SIGNAL SCALING:

Software-based amplification will enable waveform zooming for low-amplitude signals.

1.3.5 EDUCATIONAL AND PRACTICAL APPLICATION:

The project is expected to serve as a learning tool for understanding signal sampling, ADC operation, and real-time data visualization in embedded systems.

1.3.6 RELIABLE PERFORMANCE AND FLEXIBILITY:

The oscilloscope will demonstrate stable operation with the ability to handle a variety of analog input signals, making it useful for testing sensors, circuits, and function generator outputs.

1.3.7 COST-EFFECTIVE ALTERNATIVE:

The developed system will serve as an affordable replacement for conventional oscilloscopes, ideal for basic circuit testing and learning environments.

1.4 EXISTING SYSTEM

In the existing system, traditional oscilloscopes such as analog oscilloscopes and digital storage oscilloscopes (DSOs) are widely used to analyze and visualize electrical signals. These instruments display how a signal varies with time, helping engineers and researchers measure parameters like amplitude, frequency, and phase difference.

Conventional oscilloscopes provide high accuracy, fast response, and advanced features such as multi-channel input, trigger functions, and data storage. However, these devices are often expensive, bulky, and complex to operate, making them less accessible for students, hobbyists, and small laboratories. Additionally, traditional DSOs require dedicated hardware and calibration, which increases cost and maintenance requirements.

Some computer-based or USB oscilloscopes have also been developed, which connect to PCs for signal visualization. Although these are more compact and affordable than standalone oscilloscopes, they still depend on additional hardware and software setups.

Therefore, there is a need for a simple, low-cost, and portable oscilloscope system that can perform basic signal monitoring and waveform display without the limitations of traditional equipment. This limitation in the existing system motivates the development of an Arduino-based real-time oscilloscope, which offers an economical and educational alternative for basic circuit analysis and signal observation.

DISADVANTAGES

1. Limited frequency range (only low-frequency signals).
2. Lower accuracy compared to commercial oscilloscopes.
3. Single input channel only.
4. Slow display refresh rate.
5. Can measure only 0–5V signals.
6. Lacks advanced features (trigger, zoom, storage, etc.).
7. Not suitable for high-frequency or industrial applications.

1.5 PROPOSED SYSTEM

The proposed system introduces a low-cost, portable, and user-friendly real-time oscilloscope built using an Arduino microcontroller and a TFT LCD display. It is designed to overcome the high cost and complexity of conventional oscilloscopes while providing basic waveform visualization for educational and experimental purposes.

In this system, the analog input signal is fed into the Arduino's Analog-to-Digital Converter (ADC), which samples and converts it into digital data. The microcontroller then processes these values and displays them on the TFT screen as a waveform that continuously updates in real time. This setup allows users to observe voltage variations over time directly on the display.

The proposed design is compact, affordable, and easy to assemble, making it ideal for students, beginners, and small laboratories. It can be used to monitor signals from sensors, potentiometers, or small circuits. Even though it cannot fully replace professional oscilloscopes, it serves as an effective learning and testing tool for basic signal analysis and embedded system education.

This system is a low-cost alternative to commercial oscilloscopes, providing a simple and effective way to visualize electrical signals in real time. It is especially useful for students, hobbyists, and educational laboratories where affordability and simplicity are key.

ADVANTAGES

1. Low cost and affordable.
2. Portable and compact design.
3. Easy to build and operate.
4. Educational and beginner-friendly.
5. Displays signals in real time.
6. Low power consumption.
7. Open-source and customizable.
8. Useful for basic circuit and sensor testing.

1.6 TOOLS REQUIRED

1.6.1 HARDWARE COMPONENTS

1. Arduino Mega 2560:

Acts as the main controller of the system. Reads the analog input signal using its inbuilt 10-bit ADC and processes the data for real-time display. Provides sufficient memory and pins for handling both display and analog input simultaneously.

2. 2.4-inch TFT LCD Display (SPFD5408/ILI9341 Controller):

Used for graphical waveform display. Resolution: 320×240 pixels (QVGA). Operates on 5V logic and interfaces easily with Arduino Mega through parallel communication.

3. XR2206 Function Generator:

Provides the analog input waveform (sine, square, or triangular) for testing. Output voltage range should be within 0–5V to prevent damage to the Arduino input pin.

4. Connecting Wires / Jumper Cables (Male-to-Male Jumpers):

Used to make reliable connections between Arduino, display, and external components such as the button or signal input.

5. Power Supply:

Provides 9V DC power to the XR2206 Function generator.

1.6.2 SOFTWARE COMPONENTS:

1. Arduino IDE (Integrated Development Environment)2.3.6:

The primary programming environment used to write, compile, and upload code to the Arduino board. Programming Language: Arduino C/C++ (based on embedded C).

2. TFT Display Library:

A software library (such as *Adafruit_GFX* and *SPFD5408_TFT* or equivalent) used to control the LCD display and draw graphics like waveforms.

- **Adafruit GFX Library:** Purpose: Provides core graphics functions such as drawing lines, shapes, text, and pixels on the TFT display.
- **MCUFRIEND_kbv Library :** Purpose: Enables communication between the Arduino and the 2.4-inch TFT LCD (SPFD5408/ILI9341).

1.7 ORGANIZATION OF THE REPORT

- Preamble
- Literature Survey
- System Design
- Software Design
- Results and Discussion
- Advantages, Disadvantages and Applications
- Conclusion and Future scope
- References

CHAPTER 2

LITERATURE SURVEY

[1] Title: “A Low Cost Portable Oscilloscope Based on Arduino and GLCD”

Author: Ishtiak Ahmed Karim, 3rd INTERNATIONAL CONFERENCE ON INFORMATICS, ELECTRONICS & VISION 2014.

Oscilloscopes are very useful electronic instruments that enable one to see voltage wave-shapes in an electronic circuit. But these instruments are very expensive and average students can hardly afford to buy one. This paper describes the design and development of a low-cost portable oscilloscope based on Arduino and Graphical Liquid Crystal Display (GLCD). This project is designed and developed to achieve the same functionality that traditional oscilloscopes have. An Atmel microcontroller is used for data acquisition which will then be displayed as a waveform on a GLCD screen as it would have appeared on a traditional CRT oscilloscope. The oscilloscope displays waveforms in real-time by reading a finite number of samples and storing them into its internal RAM. Once the memory is full the Microcontroller will stop sampling and transfer the data to the GLCD display. This handheld oscilloscope is very cheap, uses low power, generates low internal noise, uses very few components and easy to operate.

[2] Title: “Low Cost Handheld Digital Oscilloscope”

Author: I Fushshilat and D Barmana, International Symposium on Materials and Electrical Engineering (ISMEE) 2017.

This paper discusses the design and development of handheld digital oscilloscope at low cost using embedded microcontroller system. Oscilloscope is an important instrument that must exist in every electrical and electronic laboratory, but the oscilloscope price is relatively expensive so the purpose of this paper will be describe of research to create oscilloscope using a low cost digital system. Unlike general oscilloscopes that have large dimensions, the oscilloscopes planned and built in this research will have smaller dimensions with smaller packages, so the oscilloscope will be easy to held and mobile, easy to carry anywhere, and easy to use.

[3] Title: “Portable Digital Oscilloscope using Arduino”

Author: Avanthika Pavuluri, Vardhana Jangam, S.N.V.S.K Chaitanya, Nagaraju Adimulam and Vennela Nalathoti, Proceedings of the International Conference on Inventive Computation Technologies (ICICT 2023).

The portable digital oscilloscope powered by the Atmega328p can be very helpful in overcoming these drawbacks. The advantages of a portable digital oscilloscope employing an Atmega328p are low cost, portability, light weight, low power consumption, ease of use, and others. The Atmega328p IC, an Organic Light Emitting Diodes (OLED) display, a breadboard, tactile switches, resistors, capacitors, diodes, and jumper wires are the components utilized. The importance of this research study comes from its ability to demonstrate that oscilloscopes based on the Atmega328p can be just as helpful as conventional oscilloscopes. The system will be able to function like a standard oscilloscope, which makes it useful in electronics labs and as a standalone instrument. The challenges faced by the conventional oscilloscope is that the oscilloscope cannot be carried out of the laboratories and for the oscilloscope to buy is not economical, that's where the oscilloscope using Arduino comes into the picture.

[4] Title: “An Arduino based Portable Oscilloscope”

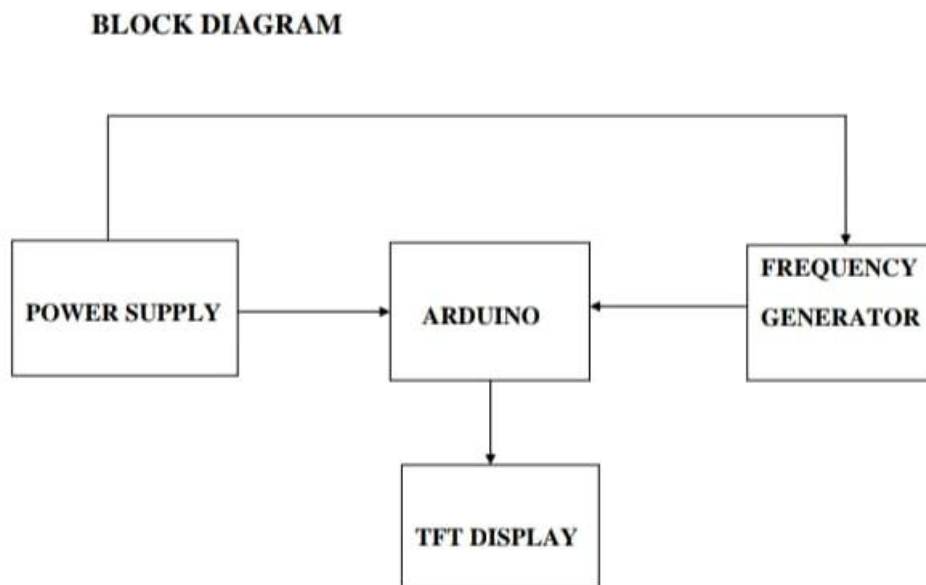
Author: Prapti Dongaonkar, Suhani Gunje, Shravani Gaikwad and Sunil Tayde, Proceedings of the 5th International Conference on Data Intelligence and Cognitive Informatics (ICDICI-2024).

This paper addresses these challenges by proposing a portable mini oscilloscope based on the Arduino NANO microcontroller, offering a more accessible and affordable alternative. Additionally, the software components responsible for signal processing, data analysis, and display rendering are discussed to highlight the oscilloscope's versatility and effectiveness in real-time signal analysis. Performance evaluations of the portable mini oscilloscope demonstrate its capabilities and limitations, emphasizing its potential applications in educational, and professional settings. The Arduino NANO microcontroller provides a versatile platform for signal analysis, making the oscilloscope suitable for a wide range of users.

CHAPTER 3

SYSTEM DESIGN

3.1 BLOCK DIAGRAM OF REAL TIME OSCILLOSCOPE



3.2 HARDWARE COMPONENTS:

3.2.1 Arduino mega 2560

The Arduino Mega 2560 is a powerful microcontroller board based on the ATmega2560 chip. It is designed for projects that require more input/output pins, memory, and processing capability than the standard Arduino Uno. It is ideal for complex electronics projects such as real-time data processing, signal monitoring, robotics, and oscilloscope applications.

Key Features

1. High Pin Count:

- Provides 54 digital pins and 16 analog inputs, allowing multiple sensors, displays, and peripherals to be connected simultaneously.
- Ideal for projects needing multiple I/O connections, like the oscilloscope's analog input and display control.

2. Enhanced Memory and Speed:

- 256 KB flash memory allows large and complex programs to be uploaded.
- 16 MHz clock ensures fast signal sampling and smooth display performance.

3. Multiple Serial Ports:

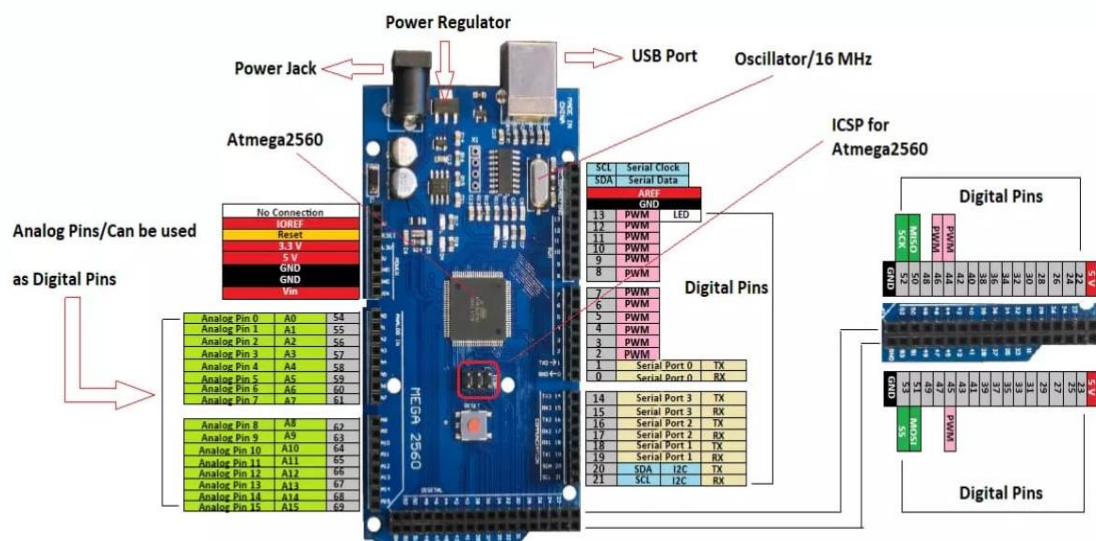
- Comes with 4 UARTs (Serial0–Serial3) for easy communication with external modules such as Bluetooth, GPS, or sensors.

4. Integrated Power Management:

- Can be powered via USB, DC adapter, or battery, with automatic power source selection.

5. Compatibility

- Works seamlessly with the Arduino IDE for easy programming.
- Supports various shields and libraries, including Adafruit_GFX and MCUFRIEND_kbv, used in your project.



Arduino Mega 2560

Fig 3.2: Arduino Mega 2560

TECHNICAL SPECIFICATIONS

Feature	Description
Microcontroller	ATmega2560
Operating Voltage	5V
Input Voltage (recommended)	7–12V
Input Voltage (limit)	6–20V
Digital I/O Pins	54 (of which 15 can be used as PWM outputs)
Analog Input Pins	16 (A0–A15)
DC Current per I/O Pin	20 mA
Flash Memory	256 KB (8 KB used by bootloader)
SRAM	8 KB
EEPROM	4 KB
Clock Speed	16 MHz
USB Connection	Type-B
Power Jack	Barrel jack (7–12V DC input)
Microcontroller Communication	UART, SPI, I ² C

Role in this Project

- The Arduino Mega 2560 reads analog input signals from the function generator through its analog pin (A15).
- The internal 10-bit ADC converts the analog waveform into digital data.
- The processed data is sent to the 2.4" TFT LCD for visualization.

- The board's high memory and multiple I/O pins make it ideal for smooth, real-time signal display.

3.2.2 XR2206 Function Generator

The XR2206 is a monolithic integrated circuit (IC) used to generate different periodic waveforms. It is widely used in low-cost laboratory, educational, and hobby electronics. It can generate sine, triangle, and square waves with good stability and low distortion.

Features:

- Generates sine, triangle, square waveforms
- Frequency range: 0.01 Hz to 1 MHz (practical ~100–500 kHz)
- Sinewave distortion typically ~0.5%
- Wide supply voltage: +10 V to +26 V
- Allows frequency modulation (FM) and amplitude modulation (AM)
- Adjustable frequency with external resistor & capacitor
- Easy waveform shaping and duty-cycle variation
- Low-power, simple circuitry

Pin Functions:

- Frequency resistance & capacitance connected to timing pins set oscillation (f)
- Wave shaping pins allow selection of sine/triangle
- Square wave available on separate output pin
- Modulation pins allow FM/AM control

Waveforms:

1. Sine wave: Low distortion (~0.5%)
2. Triangle wave: Linear ramp
3. Square wave: Digital output, good for clocking

Applications:

- Laboratory function generators
- Audio signal generation
- Sweep generators
- Sensors & VCO (voltage-controlled oscillators)
- FM/FSK circuits and Teaching electronics and testing circuits

Typical Circuit Description

A single timing capacitor and frequency-select resistor determine the output frequency.

- Waveform selection by wiring pins
- Output amplitude can be varied using resistive control
- Square output available directly
- Optional modulation input for FM/AM

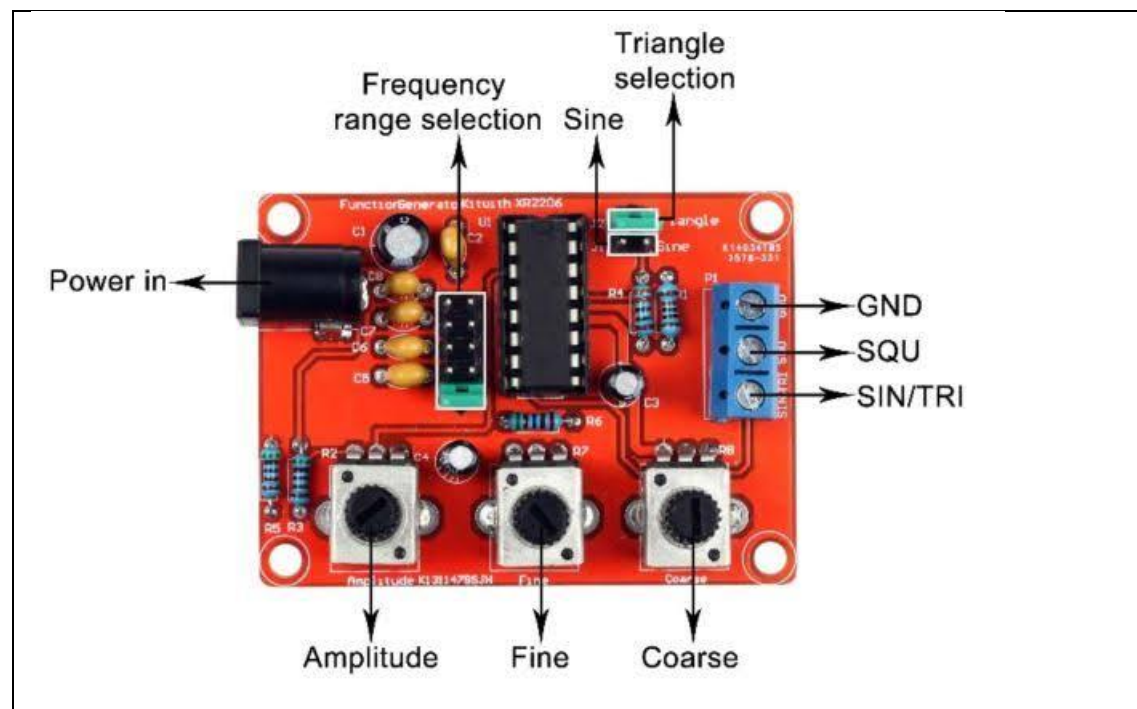


Fig 3.3: XR2206 Function Generator

3.2.3 TFT (Thin film transistor) LCD

The ILI9341 is a popular single-chip display driver IC used to control TFT LCD modules up to 240×320 resolution (QVGA). It includes a TFT controller, RAM, and display interface for microcontrollers such as Arduino, ESP32, STM32, PIC, AVR, ARM, etc.

The display uses a dedicated display driver IC called the ILI9341, which is responsible for managing pixel data, refreshing the screen, and controlling colour output. Many modules also include a touchscreen controller IC (XPT2046) to enable user input, similar to smartphones but using resistive touch technology.

Internal Functions: The IC includes:

- Display controller
- Display memory (GRAM)
- Timing generator
- Power & backlight control and Gamma correction engine

Key Features

- Supports TFT LCD panels up to 240×320 pixels
- Colour depth: 262K (18-bit) / 65K (16-bit) colors
- Built-in GRAM (Graphics RAM) = $240 \times 320 \times 18$ -bit
- Supports serial SPI and parallel interface (8/16-bit MCU)
- Wide operating voltage: 2.5V – 3.3V
- Supports sleep mode & low-power operation
- Supports display orientation (0° , 90° , 180° , 270°)

Interfaces: The ILI9341 supports multiple communication modes:

- 4-wire SPI (most common)
- 8-bit parallel MCU interface
- 16-bit parallel MCU interface
- RGB interface
- SPI is preferred in small embedded systems because it uses fewer pins.

Resolution & Color:

- Maximum resolution: 320×240 pixels
- Color: 65K/262K
- Pixel format selectable

Pin Overview (General TFT Module):

Typical pins:

- GND and VCC
- SCL / SCK (Clock)
- SDA / MOSI (Data)
- CS (Chip Select)
- DC (Data/Command)
- RST (Reset)
- LED / BL (Backlight)

ILI9341 TFT Display Module

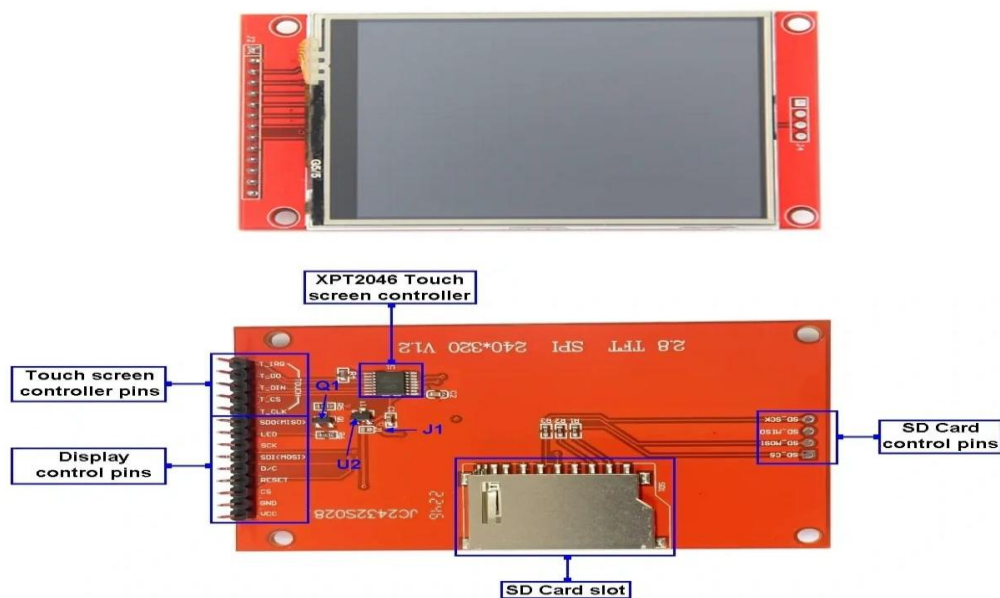


Fig 3.4: TFT Display

CHAPTER 4

MAKING OF FUNCTION GENERATOR

4.1 CIRCUIT DIAGRAM OF XR2206 FUNCTION GENERATOR

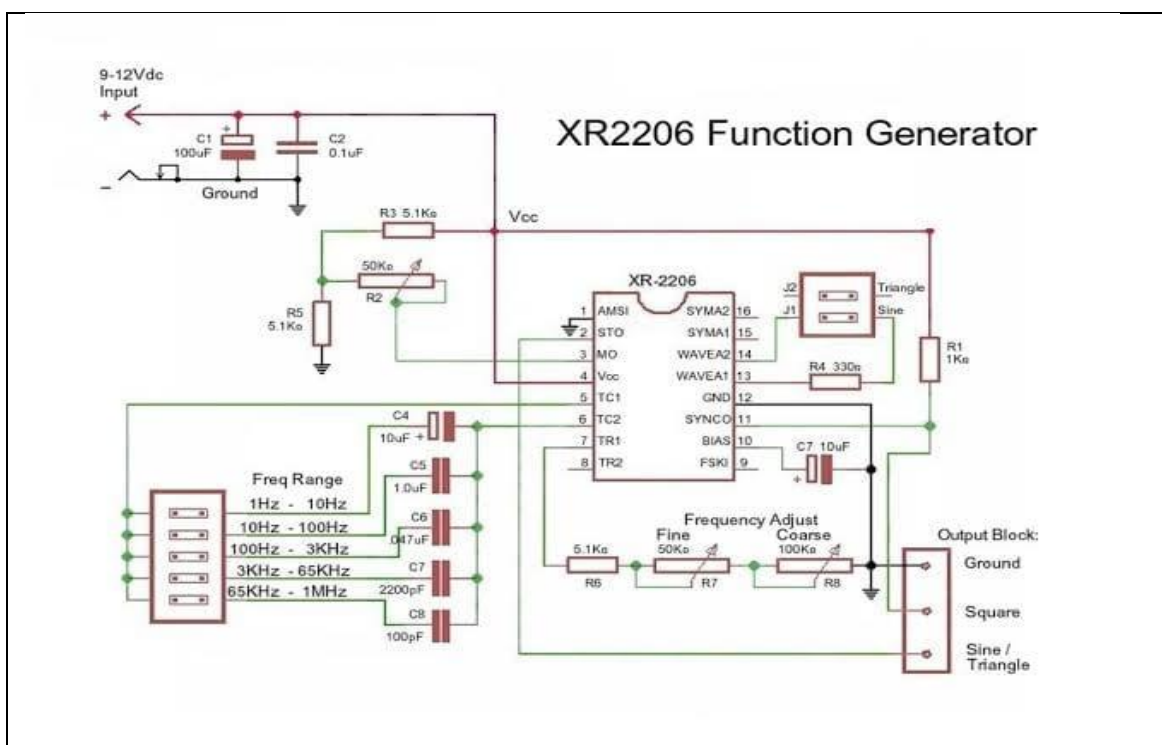


Fig 4.1: Circuit diagram of XR2206 Function Generator

Power Supply: Operates from a 9–12 V DC input, filtered by capacitors C1 (100 μ F) and C2 (0.1 μ F) for stable operation.

Main IC: The XR2206 generates different waveform types depending on external components.

Frequency Control: The output frequency is determined by timing capacitors (C4–C8) and resistors (R5–R8). A rotary switch selects the frequency range from 1 Hz to 1 MHz.

Adjustment: Two potentiometers—Fine (50 k Ω) and Coarse (100 k Ω)—allow smooth frequency variation.

Waveform Selection: Jumpers J1 and J2 select between sine and triangle wave output, while a square wave can be taken from Pin 11.

Output Section: The output block provides terminals for sine/triangle, square, and ground connections.

Components specifications of frequency generator are shown below:

Label	Type	Value	Note
R1	resistor	1K	Regardless of the polarity
R2	Potentiometer	B503=50K	(by screen printing layer)
R3, R5, R6	resistor	5.1K	Regardless of the polarity
R4	resistor	330	Regardless of the polarity
R7	Potentiometer	B503=50k	(by screen printing layer)
R8	Potentiometer	B104=100k	(by screen printing layer)
C1	Electrolytic capacitor	100UF	The positive long feet
C2	Ceramic capacitor	104	Regardless of the polarity
C3, C4	Electrolytic capacitor	10UF	The positive long feet
C5	Ceramic capacitor	105	Regardless of the polarity
C6	Ceramic capacitor	473	Regardless of the polarity
C7	Ceramic capacitor	222	Regardless of the polarity
C8	Ceramic capacitor	101	Regardless of the polarity
U1	IC(XR2206)		(by screen printing layer)
JK1	DC POWER		(by screen printing layer)
J1	2PIN Jumper cap(XM2.54)		Regardless of the polarity
J2	2PIN Jumper cap(XM2.54)		Regardless of the polarity
P1	Signal wire terminal		(by screen printing layer)
J3	2*5P Jumper cap		

Fig 4.2: Component Soldering Guide

Note

- Ceramic capacitors have no polarity (can be installed either way).
- Electrolytic capacitors C1, C2 must follow polarity and Positive = Long leg
- Potentiometers VR1, VR2 identify using printed value (B104, B503).

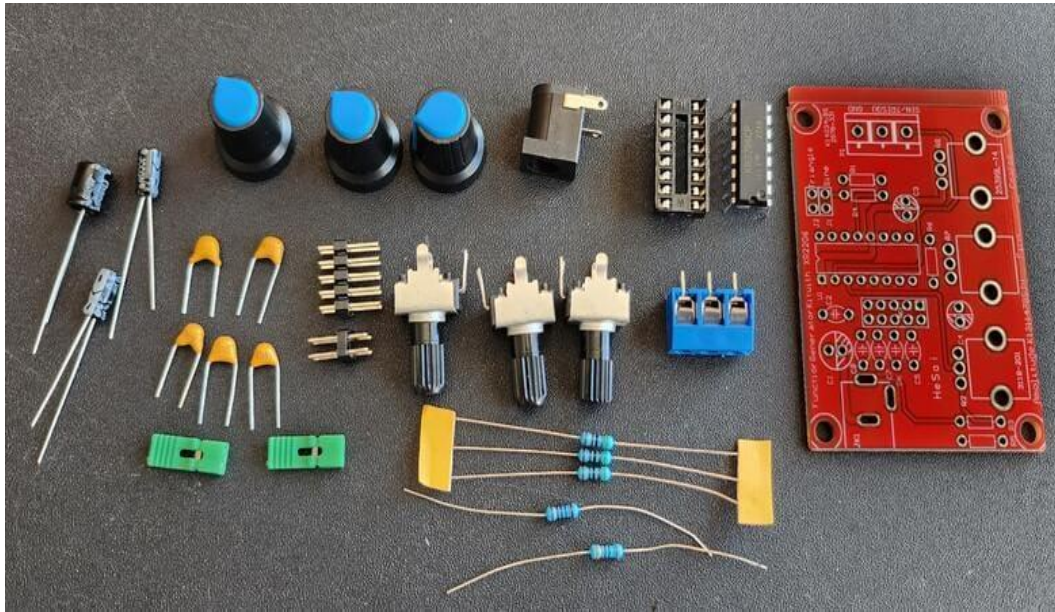


Fig 4.3: Step 1 of making Function Generator

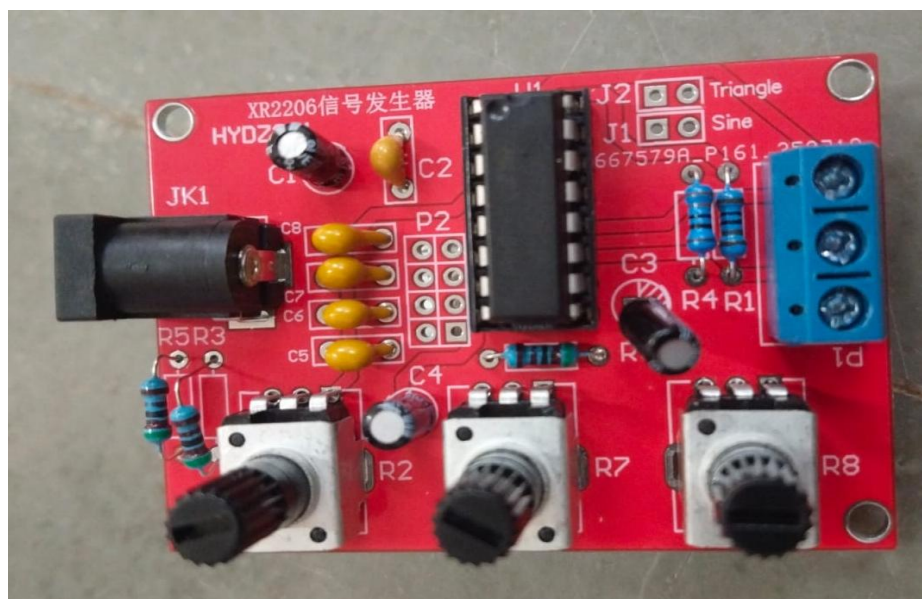


Fig 4.4: Step 2 of Making Function Generator

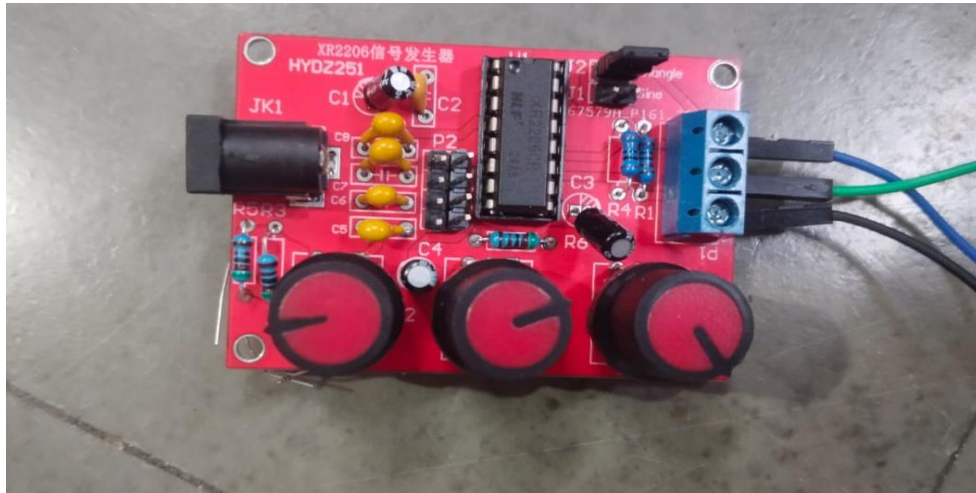


Fig 4.5: Step 3 of Making Function Generator

OUTPUT OF XR2206 FUNCTION GENERATOR

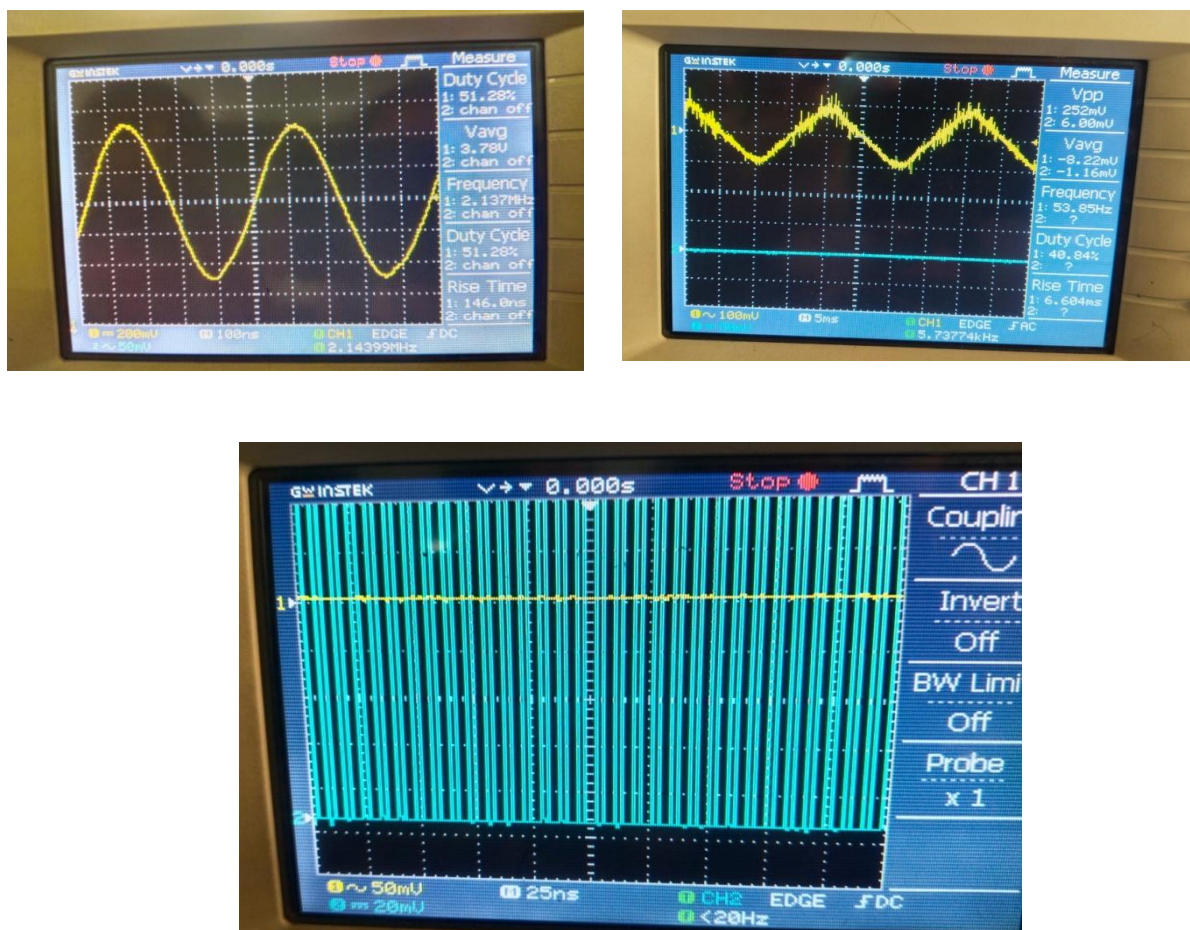


Fig.4.6: Output of XR2206 Function Generator

CHAPTER 5

HARDWARE SETUP

5.1 INTERFACING TFT DISPLAY WITH ARDUINO MEGA 2560

Physical Mounting

The TFT shield is designed to plug directly on top of the Arduino Mega 2560. No jumper wires are required because the pins are arranged to match the Mega's female headers.

Note: The same shield also fits the Arduino UNO, but the Mega uses different ports internally, which the library handles automatically.

Pin Connections

TFT Pin	Connected to Arduino Mega 2560 Pin	Function
LCD_CS	A3	Chip Select (enables display)
LCD_CD	A2	Command/Data select
LCD_WR	A1	Write signal
LCD_RD	A0	Read signal
LCD_RESET	A4	Reset the display
LCD_D0–D7	Digital 22–29	8-bit data lines
5V	5V	Power supply
GND	GND	Ground connection
LED	3.3V or 5V (through 100 Ω resistor)	Backlight power

Library Requirements

Install the following libraries from Arduino IDE → Sketch → Include Library → Manage Libraries:

1. **Adafruit_GFX:** (Core graphics library for text, shapes, and colors)
2. **MCUFRIEND_kbv :** (Driver for 2.4" TFT shields using ILI9341, SPFD5408, etc.)

Working Principle

- The Arduino Mega sends parallel data (8 bits at a time) to the TFT.
- The control pins (CS, CD, WR, RD) manage when and how the data is written.
- The Adafruit_GFX library handles text, drawing, and graphics, while MCUFRIEND_kbv manages low-level display communication.
- The display shows graphical output, such as the waveform in your oscilloscope project.

Testing

To verify the connection:

1. Open File → Examples → MCUFRIEND_kbv → graphicstest_kbv.
2. Upload the sketch to Arduino Mega 2560.
3. If wiring and libraries are correct, various colors, shapes, and text will appear.

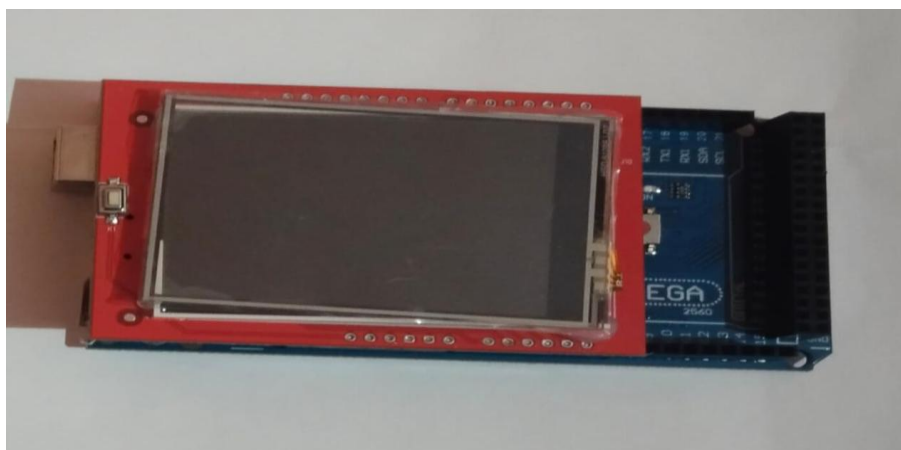


Fig: 5.1: Interfacing TFT with Arduino mega 2560

5.2 HARDWARE SETUP

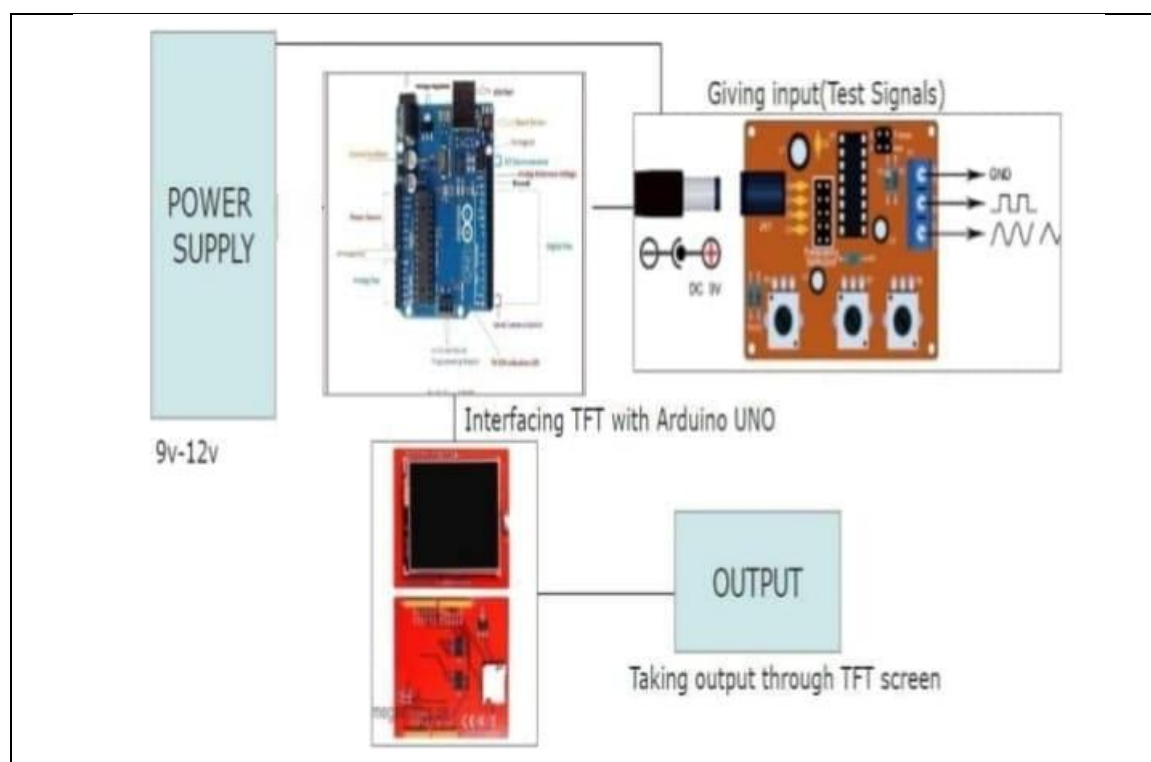


Fig: 5.2: Hardware Setup

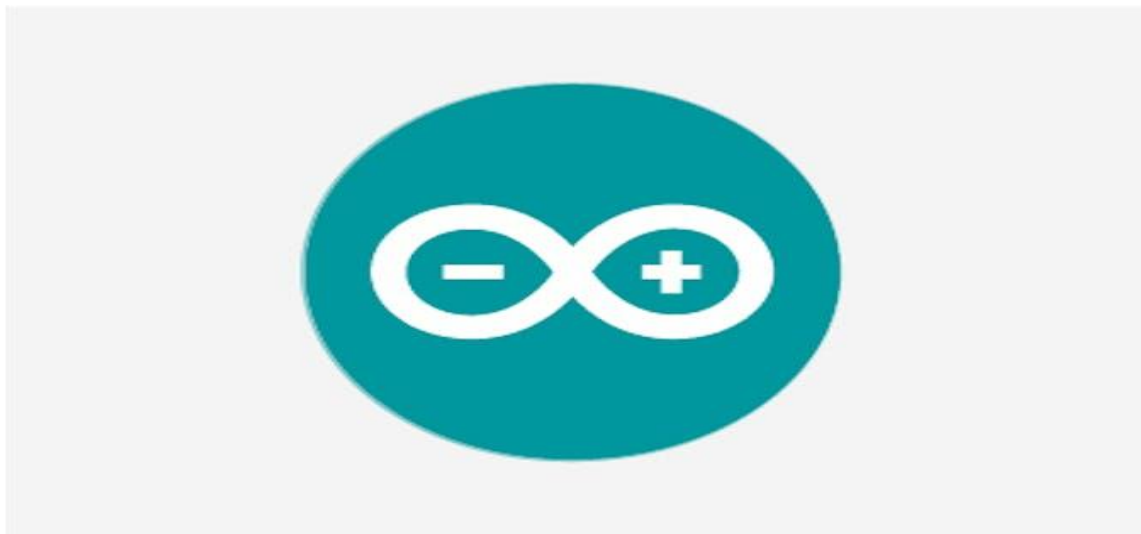
The diagram represents the working concept of a real-time oscilloscope using an Arduino-based system. The project consists of four main parts — power supply, Arduino board, function generator, and TFT display output. The power supply block provides 9V–12V DC to the function generator circuit. This stable DC input allows the function generator to produce test signals such as sine, square, or triangular waves. The output signal from the function generator is then given to the Arduino's analog input pin. The Arduino board (in your case, Arduino Mega 2560) acts as the main controller. It reads the input analog voltage using its Analog-to-Digital Converter (ADC). The Arduino converts these voltage samples into digital data points. These points are then sent to the TFT LCD display for real-time waveform plotting. The TFT screen is connected to the Arduino and acts as the output device. It shows the input waveform graphically as voltage vs. time, similar to a digital oscilloscope. The Arduino and TFT share a 5V supply (from USB or Arduino board itself). The function generator operates independently using its own 9V DC source. The ground (GND) of both modules is connected together for a common voltage reference. Overall, this setup helps observe and analyze analog signals visually through the TFT output screen.

CHAPTER 6

SOFTWARE DESIGN

6.1 ARDUINO IDE 2.3.6

Arduino IDE 2.3.6 is the latest improved version of the Arduino Integrated Development Environment. It is designed to provide a faster, smoother, and more user-friendly experience for programming Arduino boards.



Key Features:

1. Modern User Interface

- Clean and organized layout.
- Easier navigation for beginners.
- Dark mode / Light mode support for user comfort.

2. Built-in Code Editor

- Supports syntax highlighting, auto-indentation, and code folding.
- Auto-completion suggests functions and variable names.
- Shows errors in code instantly while typing.

3. Faster Compilation and Uploading

- Uses improved compiler tools and Reduces program compile time.
- Uploading sketch to Arduino board is faster compared to old IDE.

4. Board Manager and Library Manager

- Allows installing and updating boards (Arduino, ESP, STM, etc.).
- Library Manager provides thousands of libraries.
- One-click download and update support.

5. Serial Monitor and Serial Plotter

- Integrated Serial Monitor for debugging.
- Serial Plotter shows real-time sensor waveforms.
- Helps in testing and debugging without extra tools.

6. Multiple File and Tabs Support

- Allows multi-file projects.
- Useful for large projects with many program modules.

7. Extensions / Plugin Support

- Allows installing plugins and language support tools.
- Example: C/C++ IntelliSense, debugging extensions, etc.

8. Cross-Platform Compatibility

Works on:

- Windows and Linux
- macOS

9. Integrated Debugging (for supported boards)

- Step-by-step debugging for boards that support debugging.
- Helps identify logical and runtime errors.

10. Cloud Sketch Sync (Arduino Cloud Support)

- Allows syncing code with Arduino IoT Cloud.
- Useful for online storage and remote project access.

11. Portable and Offline Mode

- Can be used without internet after installation.
- Suitable for lab and classroom environments.

Libraries and Examples

- Libraries: Extend the functionality of the IDE with pre-written code for sensors, actuators, and modules. You can install libraries directly from the IDE.
- Examples: The IDE includes a wide range of example sketches to help you get started with various functionalities.

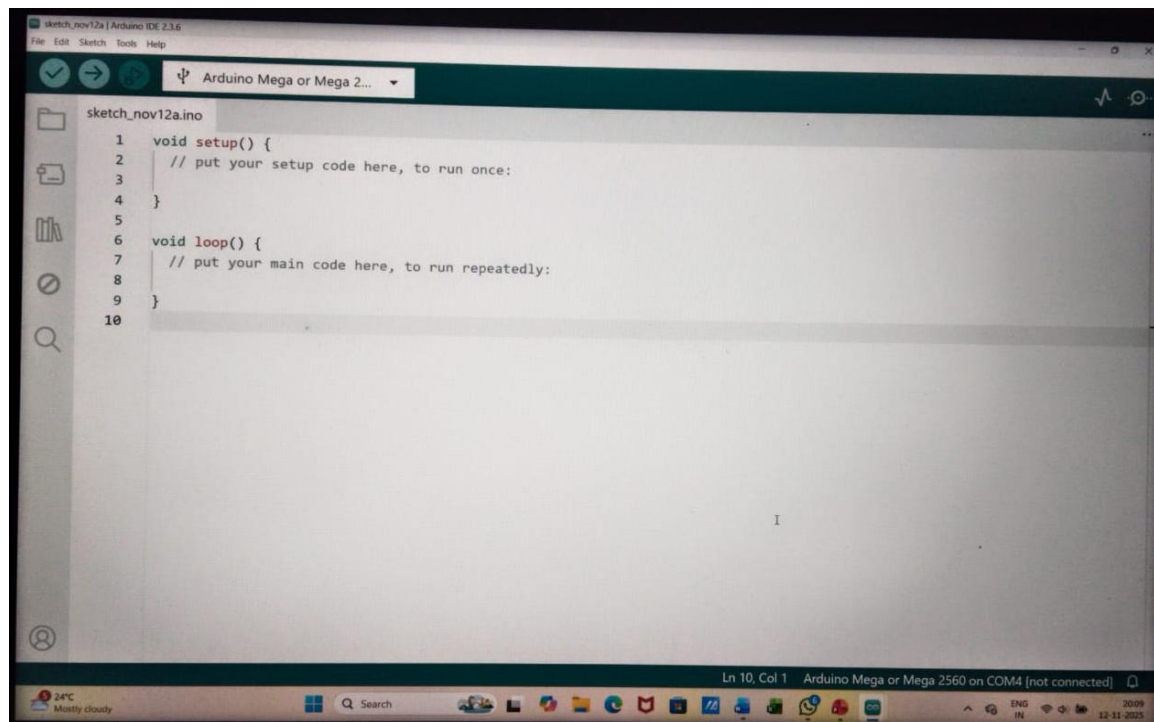


Fig:6.1 Arduino IDE Tab for Programming

6.2 CODE

```
#include <Adafruit_GFX.h>          // Core graphics library

#include <MCUFRIEND_kbv.h> // TFT driver


// TFT control pins

#define LCD_CS  A3

#define LCD_CD  A2

#define LCD_WR  A1

#define LCD_RD  A0

#define LCD_RESET A4


// Initialize display

MCUFRIEND_kbv tft(LCD_CS, LCD_CD, LCD_WR, LCD_RD, LCD_RESET);


// Analog input pin for waveform

#define ANALOG_PIN A15


// Display size

#define SCREEN_WIDTH 320

#define SCREEN_HEIGHT 240


// Colors

#define BLACK 0x0000

#define BLUE 0x001F
```

```
#define RED    0xF800

#define GREEN  0x07E0

#define CYAN   0x07FF

#define MAGENTA 0xF81F

#define YELLOW 0xFFE0

#define WHITE  0xFFFF

#define DARKGREY 0x7BEF


void drawGrid() {

    // Draw grid lines

    for (int x = 0; x < SCREEN_WIDTH; x += 20)

        tft.drawFastVLine(x, 0, SCREEN_HEIGHT, DARKGREY);

    for (int y = 0; y < SCREEN_HEIGHT; y += 20)

        tft.drawFastHLine(0, y, SCREEN_WIDTH, DARKGREY);


    // Axis labels

    tft.setTextColor(WHITE);

    tft.setTextSize(1);

    tft.setCursor(5, 5);

    tft.print("0V");

    tft.setCursor(5, SCREEN_HEIGHT - 15);

    tft.print("5V");

    tft.setCursor(SCREEN_WIDTH - 45, SCREEN_HEIGHT - 15);

    tft.print("Time ->");

}
```

```
void setup() {  
  
    Serial.begin(9600);  
  
    tft.reset();  
  
    tft.begin(0x9341);    // SPFD5408 controller  
  
    tft.setRotation(1);    // Landscape mode  
  
    tft.fillScreen(BLACK);  
  
    drawGrid();  
  
  
    tft.setTextColor(YELLOW);  
  
    tft.setTextSize(2);  
  
    tft.setCursor(30, 10);  
  
    tft.print("Real-Time Oscilloscope");  
  
    delay(1500);  
  
    tft.fillScreen(BLACK);  
  
    drawGrid();  
  
}  
  
  
void loop() {  
  
    int prevY = map(analogRead(ANALOG_PIN), 0, 1023, SCREEN_HEIGHT - 1, 0);  
  
  
    for (int x = 0; x < SCREEN_WIDTH; x++) {  
  
        int sensorValue = analogRead(ANALOG_PIN);  
  
        int y = map(sensorValue, 0, 1023, SCREEN_HEIGHT - 1, 0);
```

```
// Draw waveform

tft.drawLine(x, prevY, x + 1, y, GREEN);

prevY = y;


// Clear old trace as screen fills

if (x >= SCREEN_WIDTH - 2) {

    tft.fillScreen(BLACK);

    drawGrid();

    x = 0;

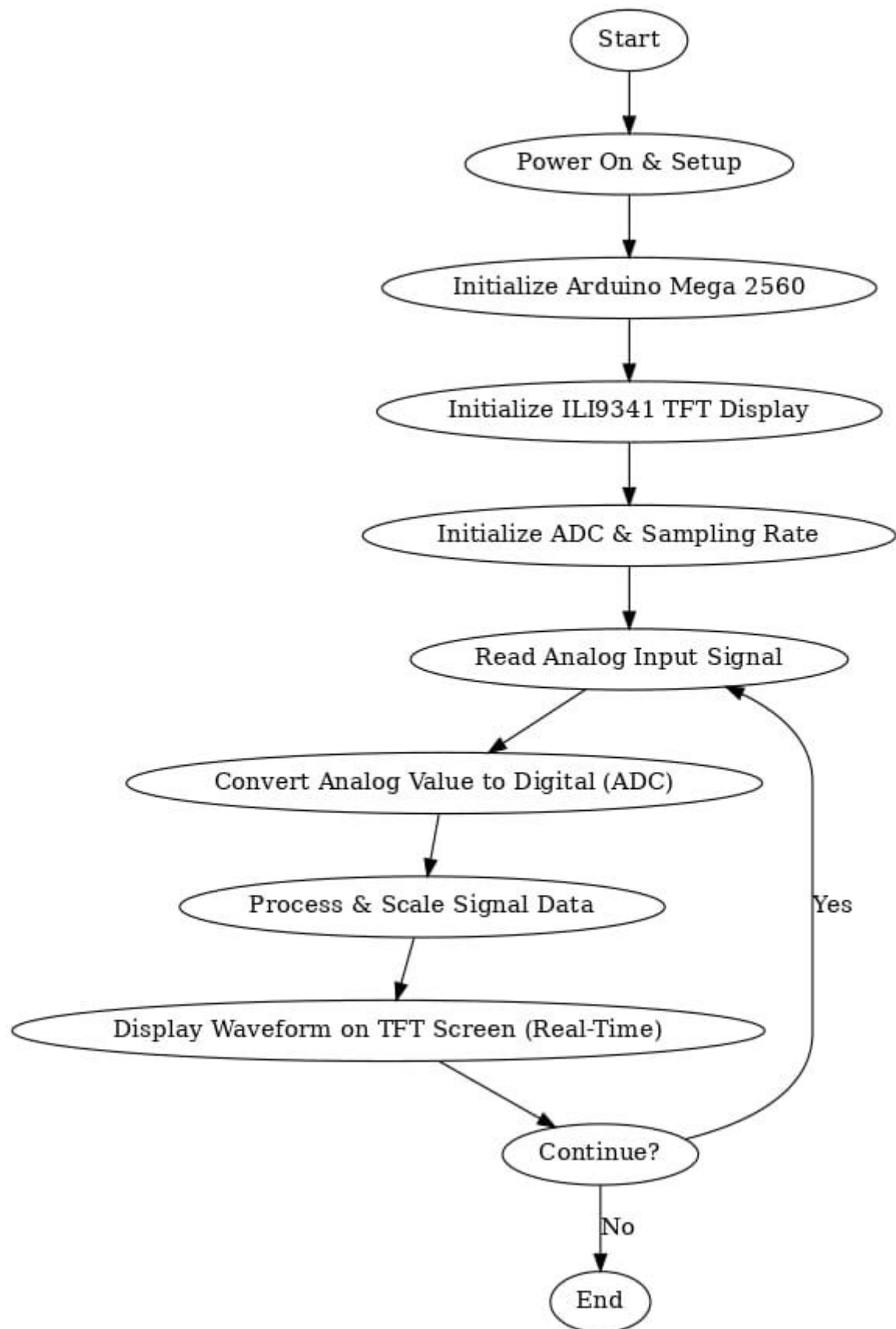
}

delay(3); // Adjust refresh rate (lower = faster)

}

}
```

6.3 FLOW CHART



CHAPTER 7

RESULTS AND DISCUSSION

7.1 THE FOLLOWING RESULTS MODEL

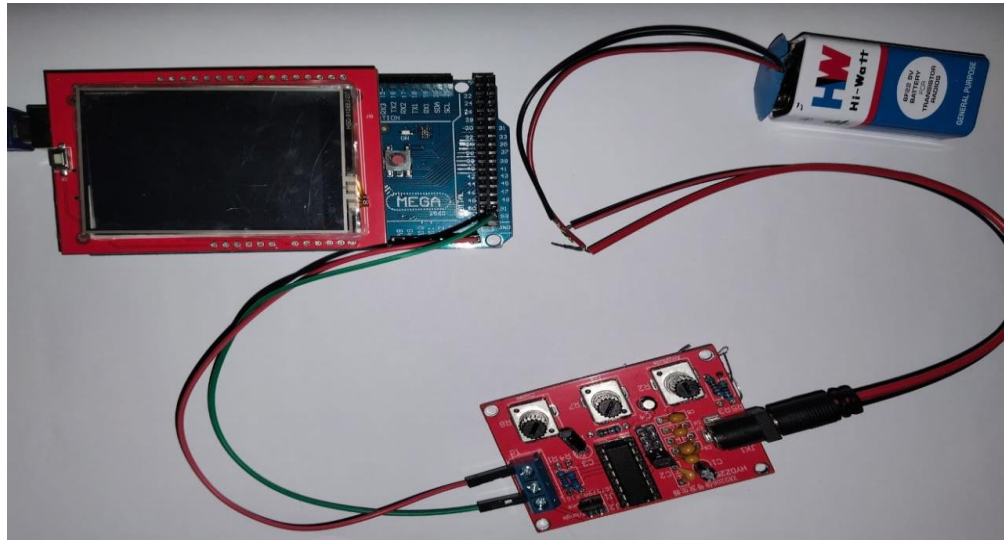


Fig:7.1 Project Setup

7.2 RESULTS

A low-cost, real-time oscilloscope built with an Arduino Mega 2560, XR2206 function generator (signal source), and a 2.4" TFT display (MCUFRIEND_kbv + Adafruit_GFX) that displays sine / triangle / square waveforms on a grid in real time.

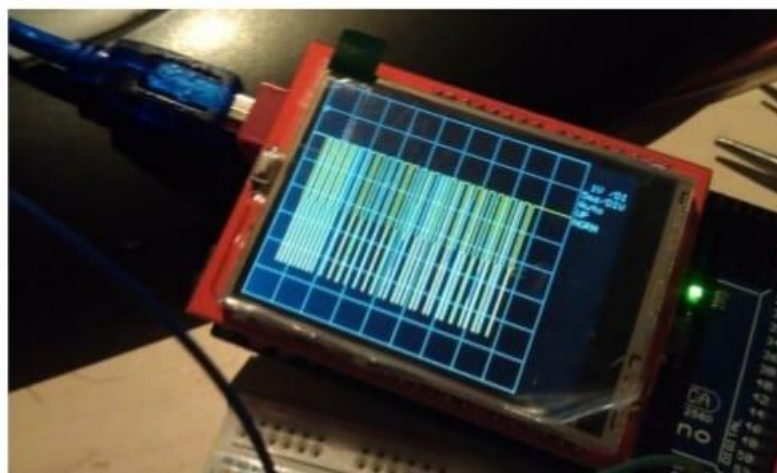


Fig:7.2 Result

7.3 DISCUSSION

The Real-Time Oscilloscope using Arduino Mega 2560 is designed to visualize electrical waveforms in real time using a low-cost microcontroller-based setup. The project successfully demonstrates how an Arduino can be interfaced with a TFT display to replicate the basic functions of a digital oscilloscope. The XR2206 function generator provided a controllable input signal (sine, square, and triangular waveforms), which was read by the Arduino's analog-to-digital converter (ADC) and plotted on the screen pixel by pixel.

During implementation, the Arduino Mega 2560 was chosen for its large memory capacity, higher number of analog inputs, and faster processing compared to the Uno. The 2.4-inch TFT display (based on the ILI9341/SPFD5408 controller) provided a suitable platform for displaying the waveform with clear color differentiation and a grid interface for better readability. The Adafruit_GFX and MCUFRIEND_kbv libraries were used to handle graphical operations effectively.

In operation, the circuit successfully displayed the analog waveforms from the XR2206 module in near real time. However, several practical challenges were encountered and resolved during testing. Initially, the display only showed a straight line, which was traced to the lack of a common ground between the XR2206 and the Arduino. Once a common reference ground was established, the waveform became visible. Another issue was that the displayed waveform was compressed or too small; this was resolved by adjusting the mapping range in the code and properly setting the amplitude on the XR2206 generator.

Because the Arduino's ADC can only sample in the 0–5V range, negative portions of the waveform were clipped. To overcome this, an AC coupling and biasing circuit can be used to shift the signal level to around 2.5V, enabling the Arduino to capture both positive and negative swings of the input wave. Although the display refresh rate and ADC sampling speed limit the system's bandwidth (typically a few hundred hertz), it is sufficient for educational and low-frequency applications.

The project highlights how microcontrollers can be utilized for signal visualization, measurement, and educational purposes without expensive equipment. While not a replacement for laboratory-grade oscilloscopes, the setup effectively demonstrates data acquisition, signal processing, and graphical output concepts.

Furthermore, the modular design allows future improvements, such as adding triggering functions, higher-speed sampling, or data logging through SD card integration.

Overall, the project successfully achieved its objective of developing a cost-effective, portable, and simple oscilloscope capable of displaying real-time analog signals. It provided valuable practical understanding of waveform generation, ADC operation, display interfacing, and embedded system design.

7.3 ADVANTAGES

7.3.1 Low Cost:

The entire system is built using inexpensive and easily available components such as the Arduino Mega, TFT display, and XR2206 signal generator—making it far cheaper than commercial digital oscilloscopes.

7.3.2 Compact and Portable:

The setup is small, lightweight, and easy to carry, allowing waveform observation anywhere without bulky laboratory instruments.

7.3.3 User-Friendly Interface:

The TFT touch screen (or color display) provides a clear graphical visualization of waveforms with voltage and time axes for easy understanding.

7.3.4 Educational Value:

The project helps students learn about analog-to-digital conversion, sampling, waveform display, and microcontroller interfacing—making it ideal for academic and training purposes.

7.3.5 Real-Time Monitoring:

Signals are captured and displayed in real time, allowing users to observe changes in voltage instantly.

7.3.6 Customizable and Expandable:

The Arduino platform allows easy modification of code for different display speeds, colors, zoom levels, and waveform types.

7.3.7 Low Power Consumption:

Operates on low DC voltage (typically 5V for Arduino and 9V for XR2206), making it energy-efficient and safe for continuous use.

7.3.8 Safe for Low-Voltage Experiments:

Designed for low-voltage signal measurements (0–5V range), minimizing risk of component damage or electrical hazards during use.

7.3.9 Versatile Signal Display:

Capable of showing various waveforms (sine, square, triangle) from function generators, sensors, or other analog sources.

7.3.10 Open Source and Easy to Maintain:

Based on open-source libraries and Arduino IDE, allowing easy code updates, debugging, and community support.

7.4 DISADVANTAGES**7.4.1 Limited Bandwidth:**

The Arduino Mega's ADC and processing speed restrict the maximum measurable signal frequency to only a few hundred hertz to a few kilohertz. It cannot analyze high-frequency signals like professional oscilloscopes.

7.4.2 Low Sampling Rate:

The default `analogRead()` function and TFT drawing delay reduce the sampling rate, which can cause waveform distortion or aliasing at higher frequencies.

7.4.3 No Negative Voltage Measurement:

The Arduino ADC can only read signals between 0 V and 5 V. Any negative portion of an input waveform is clipped unless additional biasing or AC-coupling circuitry is added.

7.4.4 No Trigger Function:

The current design continuously scrolls the waveform without synchronization, which can make the display unstable for periodic signals.

7.4.5 Limited Resolution and Accuracy:

The 10-bit ADC gives only 1024 discrete levels (≈ 4.88 mV per step), which limits precision for small signal measurements.

7.4.6 Slow Display Refresh Rate:

Updating the entire TFT screen pixel-by-pixel causes lag in the waveform display, especially at higher sampling speeds.

7.4.7 Single-Channel Input:

The project measures and displays only one signal at a time, unlike dual-channel or multi-channel professional oscilloscopes.

7.4.8 No Data Storage or Analysis:

The system does not store or analyze signals. It only visualizes them in real time. Additional SD card or memory modules are needed for recording.

7.4.9 Requires Stable Power Supply:

Fluctuations in the Arduino's or XR2206's supply voltage can affect the waveform's amplitude and accuracy.

7.4.10 Not Suitable for High-Voltage Signals:

The circuit can safely measure only low-voltage signals (0–5 V). Higher voltages require proper voltage dividers or isolation to prevent damage.

7.5 APPLICATIONS

7.5.1 Educational and Training Tool:

Used in schools, colleges, and laboratories to demonstrate signal visualization, waveform generation, and basic oscilloscope functionality for students learning electronics.

7.5.2 Basic Circuit Testing: Helpful for checking the behavior of simple analog circuits, sensors, and function generator outputs.

7.5.3 Embedded System Debugging: Engineers and hobbyists can use it to monitor voltage variations or timing in microcontroller-based projects.

7.5.4 Signal Monitoring in DIY Projects:

Can be integrated into hobby electronics projects to visualize sensor outputs, PWM signals, or audio waveforms.

7.5.5 Function Generator Output Verification:

When connected with an XR2206 or similar function generator, it helps verify generated sine, square, and triangular waveforms.

7.5.6 Sensor Output Analysis:

Useful for observing analog sensor responses (temperature, light, sound, etc.) in real time.

7.5.7 Portable Diagnostic Tool:

Due to its small size and low power consumption, it can serve as a low-cost, portable oscilloscope for basic field measurements.

7.5.8 Arduino-Based Measurement System:

Can be extended to measure other electrical parameters like frequency, RMS voltage, or peak detection with software upgrades.

7.5.9 Educational Demonstrations of ADC and Sampling Concepts:

Helps students understand how analog-to-digital conversion, sampling, and waveform plotting work in real embedded systems.

7.5.10 Prototype Testing:

Ideal for quickly checking the signal behavior in newly designed circuits during development or troubleshooting.

7.6 CONCLUSION

The Real-Time Oscilloscope using Arduino Mega 2560 and 2.4" TFT Display successfully demonstrates how an affordable and compact system can be used to visualize electrical signals. The project provides a practical understanding of how analog signals are sampled, processed, and displayed using microcontroller-based systems.

By interfacing the Arduino with the XR2206 Function Generator, different waveforms such as sine, square, and triangular waves can be observed on the TFT screen in real time. This project effectively combines hardware (signal acquisition) and software (graphical visualization) to perform the essential function of an oscilloscope.

Although it has limitations in terms of bandwidth and accuracy compared to commercial oscilloscopes, it offers a low-cost and educational alternative for students, hobbyists, and beginners in electronics and instrumentation. The system can be further enhanced with features like frequency measurement, automatic scaling, and data logging. In conclusion, this project highlights the potential of Arduino-based systems in signal analysis, embedded design, and electronics education, providing both learning and practical experimentation opportunities.

7.7 FUTURE SCOPE

7.7.1 Increased Sampling Rate:

Future versions can use high-speed ADCs or external data acquisition modules to capture higher-frequency signals with better accuracy.

7.7.2 Multi-Channel Input:

The oscilloscope can be expanded to monitor multiple analog channels simultaneously for comparing signals.

7.7.3 Automatic Scaling and Triggering:

Implementing features such as automatic voltage scaling and trigger levels will make the display more stable and user-friendly.

7.7.4 Data Storage and Analysis:

Adding an SD card module can allow saving captured waveforms for later analysis or transfer to a computer.

7.7.5 PC or Mobile Connectivity:

The system can be upgraded with Bluetooth, Wi-Fi, or USB connectivity to display waveforms on a computer or smartphone in real time.

7.7.6 Frequency and RMS Measurement:

By enhancing the software, the oscilloscope can calculate parameters like frequency, amplitude, RMS voltage, and duty cycle.

7.7.7 Touchscreen Controls:

Using the touchscreen functionality of the 2.4" TFT display can allow users to change time/div or voltage/div settings interactively.

7.7.8 Compact Hardware Design:

A custom PCB can be designed to integrate the Arduino, display, and input circuit into a portable, handheld device.

7.7.9 Advanced Signal Processing:

Incorporating digital filters and FFT (Fast Fourier Transform) can enable frequency-domain analysis of signals.

7.7.10 Integration with IoT Systems:

The oscilloscope could be connected to IoT platforms for remote monitoring of signals in smart systems or laboratories.

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